

Toward Safe and Comfortable VR-based Positive Psychotherapy

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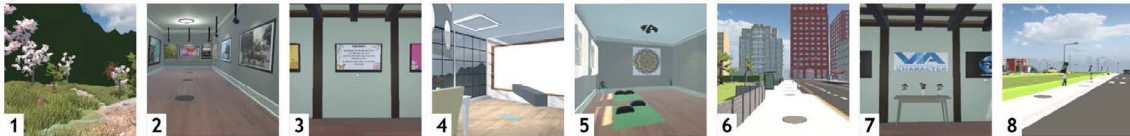


Figure 1: PositivePsychology-VR developed scenes: 1) Japanese garden; 2) Art gallery; 3) Room where users can choose to watch three videos about gratitude; 4) Room with a large whiteboard to write or draw in 3D; 5) Meditation studio; 6) City environment reflecting the daily lives of people; 7) Room where users can choose to watch six videos about personal strengths; 8) City environment with people asking questions in order to reinforce user skills.

ABSTRACT

Virtual reality has emerged as a promising tool in psychotherapy by creating immersive environments for emotional exploration. In this study, 30 participants across three sessions experienced a PositivePsychology-VR solution designed for well-being enhancement. Observations revealed minimal discomfort, with only occasional motion-related or scene-triggered anxiety that subsided over time. Participants reported increased tranquility, engagement, and emotional relief, supporting VR's potential to deepen psychotherapeutic outcomes. Minor hardware issues and pre-screening measures highlight the need for robust safety and comfort protocols. Future work will expand the participant base, refine real-time monitoring, and explore adaptive VR features for tailored psychotherapy.

Index Terms: Virtual Reality Psychotherapy, User Comfort, Positive Psychological Interventions.

1 INTRODUCTION

Virtual reality (VR) has rapidly evolved from a niche technology to a widely recognized medium for immersive experiences, allowing users to interact naturally with computer-generated elements [7]. Beyond gaming and entertainment, VR has demonstrated notable potential in clinical settings, particularly in psychotherapy, where its capacity for presence and immersion can be harnessed to improve treatment outcomes [5]. However, alongside the benefits of VR lie important considerations regarding user safety, comfort, and confidence—issues that become especially critical when addressing populations such as patients seeking mental health support. For instance, visually induced motion sickness and eye strain may prevent some individuals from tolerating head-mounted displays for extended sessions, while others might feel uneasy within immersive scenarios if certain design constraints are overlooked [9].

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In an effort to explore the role of VR in fostering positive emotional states, we previously developed a VR-based PositivePsychology software and verified its feasibility with professional psychologists in Brazil [6]. Building on that initial phase, we have now expanded our investigation to sessions involving thirty patients, capturing both quantitative and qualitative feedback on how they experienced the headset, the virtual environments, and the psychotherapeutic process overall. By employing real-world use in actual psychotherapy appointments, this extended study offers valuable insights into the relationship between VR's affordances, patient comfort, and user engagement. More specifically, it illuminates how well users adapt to multi-sensory feedback, how effectively they perceive virtual elements, and whether the new medium remains conducive to trust and well-being. Our findings underscore not only the promise of VR as a supplement to conventional psychotherapy but also the necessity of addressing safety and comfort measures head-on, ensuring that technological novelty aligns with robust clinical standards.

2 METHODOLOGY

We employed our previously developed PositivePsychology-VR application, which consists of eight virtual environments (VEs) designed to stimulate positive emotions and foster well-being. Each VE was carefully crafted to align with a specific therapeutic goal, such as gratitude, mindfulness, or personal strengths. The system ran on a Meta Quest 3 head-mounted display (HMD), featuring a 4K+ display and a guardian boundary system to enhance user safety [6]. No modifications were introduced to the original software codebase since the prior study (Figure 1). However, participants were instructed on safe use (e.g., taking breaks if needed, adjusting the head strap to manage weight distribution) and were supervised by a clinical professional throughout their VR experience.

A total of 30 patients (23 female, 7 male; ages 19–54, $M = 34$), of various occupation, for example, students, radio broadcaster and craftswoman, were recruited and pre-screened for susceptibility to VR-related adverse events, including motion sickness or significant anxiety disorders that could be aggravated by immersive stimuli. Sessions took place in a controlled setting, each patient completing three therapy sessions spaced approximately one week apart and lasting approximately one hour. After a brief introduction explaining the purpose, safe use of the VR system and talk about the topic to be worked on, patients put on the HMD and explored one VE

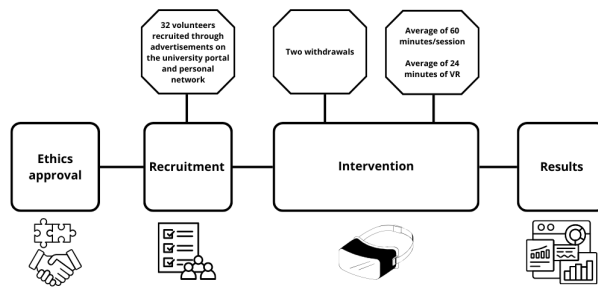


Figure 2: Complete experiment flow.

per session selected by the psychologist based on the patient's psychotherapeutic needs. Each VR exposure lasted about 24 minutes, followed by a short cool-down period and a debriefing. Throughout these sessions, the psychologist monitored participant well-being and remained alert to intervene if any discomfort arose (Figure 2).

Quantitative data were gathered via a brief comfort scale to assess any signs of motion sickness (e.g., visually induced motion sickness ratings) or physical strain. Participants also completed a short presence questionnaire to gauge immersion. Additionally, we conducted semi-structured interviews focusing on the device form factor (e.g., weight, fit), overall ease of use, and perceived safety. These qualitative discussions allowed us to capture nuanced feedback on user comfort, emotional states, and interaction experiences, providing an in-depth view of how VR-based interventions can support psychotherapy goals.

This research was approved by the Ethics Committee of the Centro Universitário Frassinetti do Recife (UNIFAFIRE), process number 79310624.4.0000.5586, ensuring full compliance with ethical standards and guidelines.

3 PRELIMINARY RESULTS

Over the course of the three therapy sessions, participants consistently reported a positive shift in their emotional states after using the VR scenarios. In the first round of sessions, some arrived feeling anxious, agitated, or simply curious, but most departed indicating greater relaxation or an overall sense of well-being. Only one individual noted mild discomfort attributable to the head-mounted display's weight, and no significant symptoms of motion sickness or eye strain were observed. During the second session, the initial mood of many participants was already more stable, with the most frequent descriptors being "tranquil" or "feeling well." After immersion, several patients described an amplified sense of calm or focus, while two others mentioned mild discomfort unrelated to the headset—one experienced fear triggered by certain content, and another felt tension in the shoulders, result of the topic worked on in the session. By the third session, no discomfort was reported, suggesting that repeated exposure may have helped participants acclimate to the virtual environment.

In terms of interaction with the eight available virtual environments, scenes 1, 5, and 7 proved most popular, collectively accounting for over half of the total usage (54/90). Scene selection was largely guided by the psychologist's assessment of each patient's psychotherapeutic needs, making it possible to tailor the VR experience to different emotional or behavioral goals, as [2] points out. Notably, patients appeared to embrace the interactive elements of the scenarios, such as writing or drawing in 3D, listening to the birds, being in nature, and reported finding these activities helpful for focusing attention away from anxious thoughts. Beyond comfort and interaction, the qualitative feedback also emphasized the positive emotional impact of the VR approach: many participants described feeling "refreshed," "calm," or "inspired," and some even

expressed gratitude for being introduced to a novel psychotherapeutic tool. Overall, these early results underline the feasibility of integrating VR into psychotherapy with manageable safety concerns, while highlighting the potential for immersive environments to deepen patient engagement and emotional processing.

4 DISCUSSION

The findings from our preliminary study reinforce the notion that VR can be integrated into psychotherapy with generally positive outcomes regarding user comfort, interaction, and emotional benefits. When viewed through the lens of mixed reality safety and interaction (MRSI), the absence of severe adverse reactions such as pronounced visually induced motion sickness or prolonged discomfort points to a growing acceptance of head-mounted displays in clinical contexts, as [4] pointed out: the availability of virtual worlds has allowed the range of interactions that would not otherwise be possible, VR is better. That said, minor reports of fear triggered by scene content and shoulder tension in the early sessions (related to the topic worked on and not to the HMD itself) underscore the critical need for vigilant monitoring, robust guardian boundaries, and flexible session protocols. Even though no major hardware-related problems arose, headset weight and potential musculoskeletal strain remain salient considerations, particularly for individuals with prolonged exposure or for kids [3] and older adults who may be more susceptible to fatigue.

From a psychological standpoint, it was encouraging to observe how rapidly most patients acclimated to the VR environment, as evidenced by the marked decline in discomfort reports by the third session, similar to [8]. This pattern suggests the potential to mitigate negative effects through session pacing, pre-session guidance, and ongoing support. However, the study's focus on a moderate sample size limits our ability to generalize results to the broader population, even though [1] validated his data with only 21 students. Moreover, participants were pre-screened to minimize severe anxiety or motion-sickness vulnerabilities, which may not fully capture how individuals with more pronounced sensitivities would respond to immersive psychotherapy. Future work will therefore need to expand the sample to diverse demographic and psychological profiles, refine safety strategies for at-risk users, and systematically evaluate motion-sickness mitigation techniques. Overall, these findings underscore both the promise and the complexity of bringing VR into psychotherapy, highlighting the need for continued refinement of comfort and safety protocols in tandem with the enrichment of psychotherapeutic efficacy.

5 CONCLUSION

The present study demonstrates that VR-based psychotherapy, using a custom PositivePsychology-VR application, can be introduced with minimal discomfort, sufficient safety measures, and notable benefits for patient engagement and emotional well-being. By assessing feedback from thirty participants over multiple sessions, we observed that initial concerns such as the headset's weight or scene-specific triggers diminished as patients grew more familiar with the immersive environment. This pattern suggests that well-planned session protocols, pre-screening for motion-sickness risk, and clear instructions on how to safely use the system can effectively mitigate many common barriers to VR adoption in clinical settings.

In moving forward, we plan to broaden our participant base to include individuals with varied backgrounds and a wider range of potential mental health issues, thereby capturing a richer spectrum of responses. Future studies will explore real-time adjustments to the virtual environment—such as changing environmental stimuli or depth cues—to better accommodate users prone to discomfort. Additionally, we aim to refine our guardian systems and integrate automated monitoring tools that detect signs of tension, motion sick-

ness, or elevated stress, which would let therapists intervene preemptively. Ultimately, by combining methodological rigor with ongoing technological improvements, we aspire to make VR an even more powerful, adaptable, and safe psychotherapeutic medium for diverse patient populations.

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